# Pet chews (including dental chews): HMTc Standard

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CORRESPONDENCE

Corrections are logged publicly on the article page, each with its date, the affected value, and the source that resolved it.

KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

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## Abstract

The Heavy Metal Tested & Certified human-grade standard for pet chews (including dental chews), stated as sold. Pet food is legally feed; HMTc voluntarily certifies each analyte to the strictest human-food sovereign maximum for the nearest analog, with the laxer feed-law floor disclosed for contrast.

## 1 | How every number on this standard is set

Pet food is legally governed by feed law — EU Directive 2002/32/EC on undesirable substances in animal feed, FDA Center for Veterinary Medicine guidance, and AAFCO model regulations — whose maximum levels are generally laxer than the corresponding human-food limits. Japan (FAMIC pet-food standard) and China (draft pet-food guidance) set enforced or proposed pet-food maxima in the same lax range. The HMTc standard for each analyte is not the feed limit: it is the strictest applicable human-food sovereign maximum for the nearest human-food analog, adopted as a voluntary human-grade over-compliance. The feed floor is disclosed on every row and never selects the number.

## 2 | Master limit table

All values in ppb (µg/kg), as sold basis. A brand meeting every value qualifies for the mark.

| ANALYTE | HMTC LIMIT | BASIS |
|---|---|---|
| Lead (Pb) | 20 | read-across (Hong Kong) |

| | | |
|---|---|---|
| Cadmium (Cd) | 40 | read-across (EU 2023/915) |
| Total arsenic (tAs) | 100 | read-across (Paraguay) |
| Inorganic arsenic (iAs) | 100 | read-across (Israel) |
| Mercury, total (tHg) | 20 | read-across (Hong Kong) |
| Methylmercury (MeHg) | ≤ tHg | bounded by tHg (reflex speciation) |
| Nickel (Ni) | 800 | read-across (EU 2024/1987) |
| Tin (Sn) | 250,000 | read-across (Singapore) |
| Aluminium (Al) | 100,000 | read-across (China) |
| Total chromium (Cr) | 500 | read-across (Hong Kong) |
| Hexavalent chromium (Cr-VI) | ≤ Cr | bounded by Cr (reflex speciation) |

## 3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

| ANALYTE | HMTC | SOVEREIGN ANALOG / DETERMINATION |
|---|---|---|
| Lead (Pb) | 20 | read-across (Hong Kong); Cereal-based food intended to be consumed principally by persons under the age of 36 months |
| Cadmium (Cd) | 40 | read-across (EU 2023/915); pulses, general |
| Total arsenic (tAs) | 100 | read-across (Paraguay); pulses, dried legume seeds, except soybean |
| Inorganic arsenic (iAs) | 100 | read-across (Israel); Dried legumes, including peanuts and soybeans |
| Mercury, total (tHg) | 20 | read-across (Hong Kong); Rice, husked rice, polished rice, maize, maize flour, wheat, wheat flour |
| Methylmercury (MeHg) | ≤ tHg | bounded by tHg (reflex speciation) |
| Nickel (Ni) | 800 | read-across (EU 2024/1987); cereals except durum/rice/husked-rice/-pseudocereal/oat rows |
| Tin (Sn) | 250,000 | read-across (Singapore); any food |

| | | |
|---|---|---|
| Aluminium (Al) | 100,000 | read-across (China); baked foods (bread, pastries, cakes, biscuits, cookies, crackers); deep-fried flour products; batter, breading, and frying powder |
| Total chromium (Cr) | 500 | read-across (Hong Kong); Vegetables, other than pulses |
| Hexavalent chromium (Cr-VI) | ≤ Cr | bounded by Cr (reflex speciation) |

## 4 | Cite this standard

Pendergrass, K. (2026). *Pet chews (including dental chews): HMTc standard* (Standard No. 10.83138/hmtc.std.17.pet-chews-dental). Institute of Contaminant Standards. https://doi.org/10.83138/hmtc.std.17.pet-chews-dental

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## References

[1] Hong Kong
Binds Pb, tHg, Cr.
READ-ACROSS

[2] EU 2023/915
Binds Cd.
READ-ACROSS

[3] Paraguay
Binds tAs.
READ-ACROSS

[4] EU 2024/1987
Binds Ni.
READ-ACROSS

[5] Singapore
Binds Sn. any food.
READ-ACROSS

[6] China
Binds Al. baked foods (bread, pastries, cakes, biscuits, cookies, crackers); deep-fried flour products; batter, breading, and frying powder.
READ-ACROSS

[7] Israel
Binds iAs.
READ-ACROSS

### STATUS AND LICENCE

A normative certification standard, editorially reviewed and version-controlled in-house; every value traces to a named sovereign maximum level, disclosed in the derivation. © 2026

### COMPETING INTERESTS

This standard is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government

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## DATA AVAILABILITY

The works this standard draws on are listed in full in the references above and on the standard page at https://heavymetalcertified.com/standards/pet-chews-dental. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

## PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/pet-chews-dental.